

Probabilistic models 1

Exercise 11

Model parameters probability distribution estimation

We want model of probability distribution $p(x)$.

It can be specified by some parametric description with parameters θ : $p_{\theta}(x)$

Model parameters have again some probability distribution given measured dataset D : $p(\theta|D)$

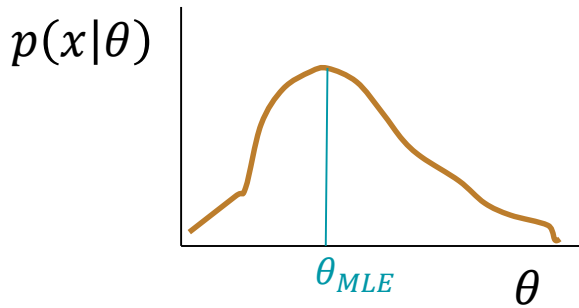
$(p(\theta|x))$ will be used in this presentation)

If we know $p(\theta|D)$, we can select most probable model parameters and use them for in $p_{\theta}(x)$ model.

Parameter estimation methods

$$p(\theta|x) = \frac{p(x|\theta) \cdot p(\theta)}{p(x)}$$

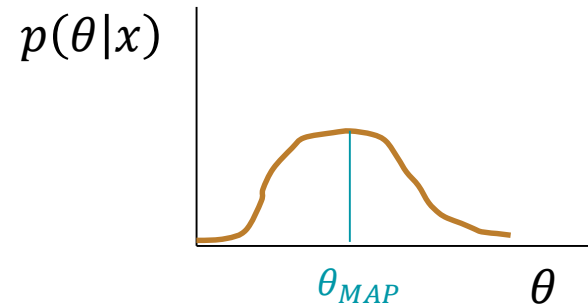
Maximum likelihood estimation
MLE



Under the assumed statistical model the observed data are most probable.

$$\operatorname{argmax}_{\theta} p(x|\theta)$$

Maximum a-posteriori probability
MAP



Find most probable model parameters for given data.

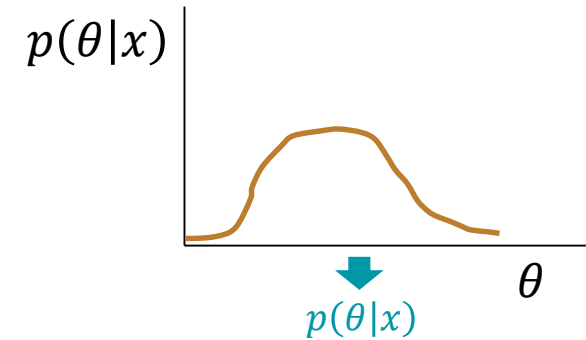
$$\operatorname{argmax}_{\theta} p(\theta|x) =$$

$$= \operatorname{argmax}_{\theta} \frac{p(x|\theta) p(\theta)}{p(x)} =$$

$$= \operatorname{argmax}_{\theta} p(x|\theta) p(\theta)$$

constant
can be
neglected for
argmax

Bayesian inference



Find model parameters probability distribution for given data.

$$p(\theta|x) =$$

$$= \frac{p(x|\theta) p(\theta)}{p(x)} =$$

$$= \frac{p(x|\theta) p(\theta)}{\int_{\Theta} p(x|\theta) p(\theta) d\theta}$$

hard to
calculate

Bernoulli model estimation

$$f(x|\theta) = \theta^x(1 - \theta)^{1-x}$$

Maximum likelihood estimation
MLE

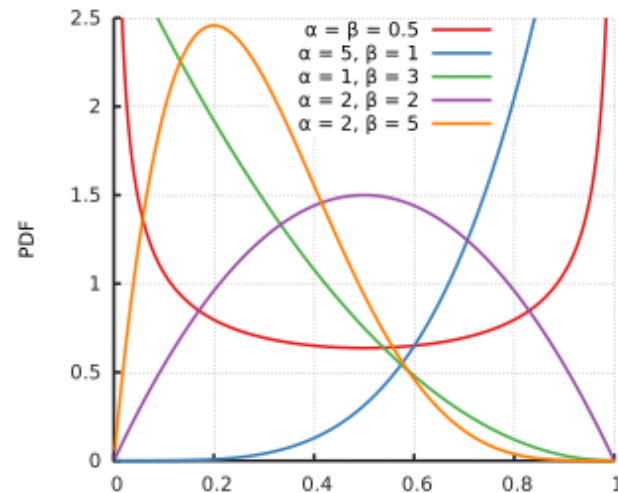
$$p(x_1 \dots, x_n, |\theta) = \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i}$$

$$\hat{\theta}_{MLE} =$$

Maximum a-posteriori probability
MAP

$$p(x_1 \dots, x_n, |\theta) = \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i}$$

$$p_{beta}(\theta | \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{1-\alpha}(1 - \theta)^{1-\beta}$$



$$\hat{\theta}_{MAP} = \frac{\sum_{i=1}^n x_i + \alpha - 1}{n + \beta + \alpha - 2}$$

Bayesian inference

“ $Beta(\alpha', \beta') = Bernoulli(\theta) Beta(\alpha, \beta)$ ”

$$p(x_1 \dots, x_n, |\theta) = \prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i}$$

$$p_{beta}(\theta | \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{1-\alpha}(1 - \theta)^{1-\beta}$$

$$\alpha' = \alpha + \sum_{i=1}^n x_i$$

$$\beta' = \beta + n - \sum_{i=1}^n x_i$$

Gaussian model estimation

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Maximum likelihood estimation

MLE

$$p(x_1 \dots, x_n, |\mu, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$\log p(x_1 \dots, x_n | \mu, \sigma) = \sum_{i=1}^n -\log(\sqrt{2\pi\sigma^2}) - \frac{1}{2\sigma^2} (x_i - \mu)^2$$

$$\frac{d \log p(x_1 \dots, x_n | \mu, \sigma)}{d \mu} = \sum_{i=1}^n 0 + \frac{1}{\sigma^2} (x_i - \mu) = 0$$

$$\hat{\mu} = \sum_{i=1}^n \frac{x_i}{n}$$

$$\hat{\mu}_{MLE} = \sum_{i=1}^n \frac{x_i}{n} = \bar{x}$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Maximum a-posteriori probability

MAP

$$p(x_1 \dots, x_n, |\mu; \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$p(\mu | \mu_0, \sigma_0^2) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \cdot \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)$$

$$p(\mu | x; \sigma^2, \mu_0, \sigma_0^2) = p(x_1 \dots, x_n, |\mu; \sigma^2) \cdot p(\mu | \mu_0, \sigma_0^2)$$

$$\hat{\mu}_{MAP} = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \cdot \mu_0$$

Bayesian inference

$$“Normal(\mu', \sigma') = Normal(\mu; \sigma) Normal(\mu_0, \sigma_0)”$$

$$p(x_1 \dots, x_n, |\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$p(\mu | \mu_0, \sigma_0^2) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \cdot \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)$$

$$p(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

$$\mu' = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \cdot \mu_0 = \mu_{MAP}$$

$$\sigma^{2'} = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}\right)^{-1}$$

MLE - numerical solution

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

If there is no analytical solution, we can still find the solution numerically.

Which leads to optimization problem:

$$\underset{\theta}{\text{minimize}} \quad -\log p(x_1 \dots x_n | \theta) = -\sum_{i=1}^n \log \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

(logarithm - numerical stability)

(minimize minus - optimization convention)

How to solve this? - optimization

Example with grid-search:

